Aaron (Jiaxun) Li

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Education

University of California, Berkeley — Aug. 2025 – May 2029 (Expected)
Ph.D. in Computer Science — GPA: 4.00/4.00
Advisors: Prof. Ion Stoica and Prof. Bin Yu

Harvard University — Sep. 2023 – May 2025
M.E. in Computational Science and Engineering (Thesis Track) — GPA: 3.91/4.00
Cross-registered at MIT EECS

University of California, Berkeley — Aug. 2019 – May 2023
B.A. in Computer Science (EECS Honors) — GPA: 3.92/4.00
B.A. in Psychology — GPA: 3.90/4.00

Research Interests

Agentic Systems, Large Language Models, Test-time Scaling, Post-training, AI Safety

Publications

*Equal contribution

[1] **BenchEvolver: Frontier Task Synthesis via Solution-Centric Evolution**
Yangzhen Wu*, **Aaron J. Li***, Wenjie Ma, Li Cao, Ziheng Zhou, Mert Cemri, Shu Liu, Yuran Xiu, Chenxiao Yan, Haikun Zhao, Bin Yu, Ion Stoica, and Dawn Song
Advances in Neural Information Processing Systems (NeurIPS) 2026

[2] **DualEval: Joint Model-Item Calibration for Unified LLM Evaluation**
Aaron J. Li, Hao Huang, Youngmin Park, Yitong Ma, Wei-Lin Chiang, Li Chen, Cho-Jui Hsieh, Bin Yu, and Ion Stoica
Findings of the Association for Computational Linguistics: EMNLP 2026

[3] **Evaluating Adversarial Robustness of Concept Representations in Sparse Autoencoders**
Aaron J. Li, Suraj Srinivas, Usha Bhalla, and Himabindu Lakkaraju
Conference of the European Chapter of the Association for Computational Linguistics (EACL) 2026

[4] **More RLHF, More Trust? On the Impact of Preference Alignment on Trustworthiness**
Aaron J. Li, Satyapriya Krishna, and Himabindu Lakkaraju
International Conference on Learning Representations (ICLR) 2025 **(Oral Presentation, Top 1.8%)**

[5] **Improving Prototypical Visual Explanations with Reward Reweighing, Reselection, and Retraining**
Aaron J. Li, Robin Netzorg, Zhihan Cheng, Zhuoqin Zhang, and Bin Yu
International Conference on Machine Learning (ICML) 2024

[6] **Certifying LLM Safety against Adversarial Prompting**
Aounon Kumar, Chirag Agarwal, Suraj Srinivas, **Aaron J. Li**, Soheil Feizi, and Himabindu Lakkaraju
Conference on Language Modeling (COLM) 2024

Preprints

[1] **Green Shielding: A User-Centric Approach Towards Trustworthy AI**
Aaron J. Li, Nicolas Sanchez, Hao Huang, Ruijiang Dong, Jaskaran Bains, Katrin Jaradeh, Zhen Xiang, Bo Li, Feng Liu, Aaron Kornblith, and Bin Yu
arXiv preprint: 2604.24700

[2] **Speaker Disentanglement of Speech Pre-trained Model Based on Interpretability**
Junhua Li, Xiaoxu Zhu, and **Aaron J. Li**
arXiv preprint: 2507.17851

Research Experience

University of California, Berkeley Berkeley, CA
Graduate Student Researcher, advised by Prof. Ion Stoica and Prof. Bin Yu *Aug. 2025 – Present*
Sky Computing Lab & Berkeley AI Research (BAIR)

- Developed open-source frameworks for generalizable, end-to-end cost-saving strategies for frontier coding agents.
- Developed a novel data generation framework for synthesizing high-quality frontier coding tasks that challenge state-of-the-art LLMs without relying on external verifiers.
- Proposed a scalable LLM evaluation framework for realistic medical diagnosis tasks without objective ground truth; conducted case studies on how common user biases influence model behavior.

Harvard University Cambridge, MA
Graduate Student Researcher, advised by Prof. Hima Lakkaraju *Sep. 2023 – May 2025*

- Conducted the first systematic evaluation of RLHF's impact on trustworthiness, revealing conflicts between alignment goals and dataset limitations.
- Developed a novel evaluation framework for adversarial robustness of sparse autoencoders, revealing inherent limitations in their concept space.
- Provided certified robustness guarantees for empirical defenses against adversarial prompting of LLMs.

University of California, Berkeley Berkeley, CA
Undergraduate Researcher, supervised by Prof. Bin Yu *Aug. 2022 – May 2023*

- Improved the interpretability of prototype-based CNNs by aligning generated visual explanations with collected human preferences.

Industry Experience

Microsoft Research Redmond, WA
Research Intern, Deep Learning Group *May 2026 – Aug. 2026*

- Improved token efficiency of proprietary coding agents through automatic, experience-driven adaptation of repository-specific knowledge.

Arena Intelligence Inc. San Francisco, CA
Student Researcher *Mar. 2026 – Present*

- Proposed a unified framework of benchmark- and preference-based evaluation signals that jointly estimates task difficulties and LLM capabilities on a shared latent scale.

Shanghai AI Lab Shanghai, China
Research Intern, Speech Group *Jun. 2023 – Aug. 2023*

- Explored interpretability methods to disentangle speaker and content information in representations.

Teaching Experience

Course Staff, UC Berkeley EECS

- CS 170: Efficient Algorithms and Intractable Problems Fall 2021
- CS 188: Introduction to Artificial Intelligence Summer 2021
- CS 70: Discrete Mathematics and Probability Theory Summer 2020

Skills

Programming Python, C++, Java, C, MATLAB, R
Frameworks PyTorch, vLLM, SGLang, CUDA, Ray, TensorFlow, Linux
Languages English (Fluent), Chinese (Native)

Coursework

Graduate: Natural Language Processing, AI for Systems, Inverse Reinforcement Learning, Sensorimotor Learning, Spoken Language Processing, Geometric Machine Learning, Efficient Machine Learning

Undergraduate: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Probability and Random Processes, Convex Optimization, Signal Processing, Efficient Algorithms, Human Neuroanatomy, Neuroimaging, Computational Models of Cognition